2. Asiatic rice borer (*Chilo suppressalis*)

Last modified: October 29, 2026

2A. Introduction

The Asiatic rice borer (ASRB), *Chilo suppressalis* Walker (Lepidoptera: Crambidae), is a stem borer and major pest of rice and other grasses, including water oat, sorghum, millet, and corn, throughout Asia, Oceania, the Middle East, and Europe (CAPS 2025, Meng et al. 2008). ASRB larvae cause major damage to rice crops by severing panicles or the vascular system of tillers, leading to white earheads and dead hearts (Kiritani and Iwao 1967, Muralidharan and Pasalu 2006, Chen et al. 2014, EPPO 2019). Larvae can cause substantial economic losses, with reports of 20% yield loss under some conditions (Rahman et al. 2004). The pest is present in Hawaii and is frequently intercepted at United States ports of entry (Solis 2006). With the United States rice industry valued at over USD 6 billion in 2025 (Childs 2025), yield reductions resulting from ASRB could cause significant economic damage to rice-producing regions.

A model for ASRB developed for the Degree-Day, establishment Risk, and Phenological event mapping system (DDRP) is operationalized to produce near real-time forecasts for the contiguous United States (CONUS) at https://uspest.org/DDRP. The full set of model parameters for ASRB, including developmental temperature thresholds, degree-day requirements for individual life stages, and climate stress thresholds and limits, are available at USPest.org (Coop and Barker 2020).

2B. Results

i. Phenology

Entire period: 1981–2025

Trends in predicted first adult emergence and egg hatch for ASRB over the 44-year period (1981–2025) were predominantly negative, indicating that both phenological events shifted earlier over time. Earlier phenology was spatially extensive, occurring across much of the CONUS, although trends toward later phenology occurred in portions of the northern interior and western United States (Figure 1, Figure 2). Statistically significant trends were more spatially restricted, occurring primarily in the East, the South, and higher elevations of the West. Within these regions, predicted adult emergence and egg hatch shifted earlier by approximately 0.01–0.5 days per year, with some of the largest shifts occurring at higher elevations in the West. Only a few small, isolated areas exhibited significant trends towards later phenology.

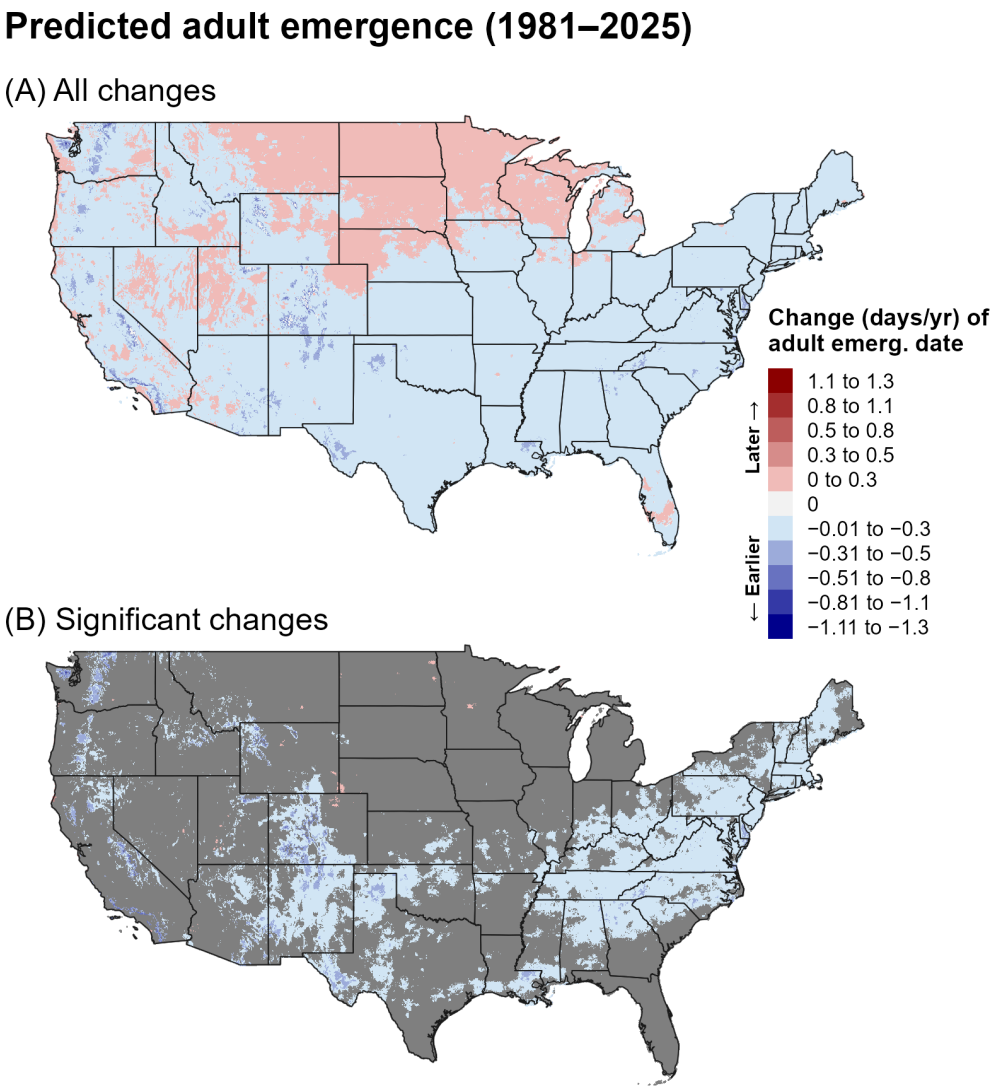


Figure 1. Changes (days/year) in predicted dates of first adult emergence for ASRB across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

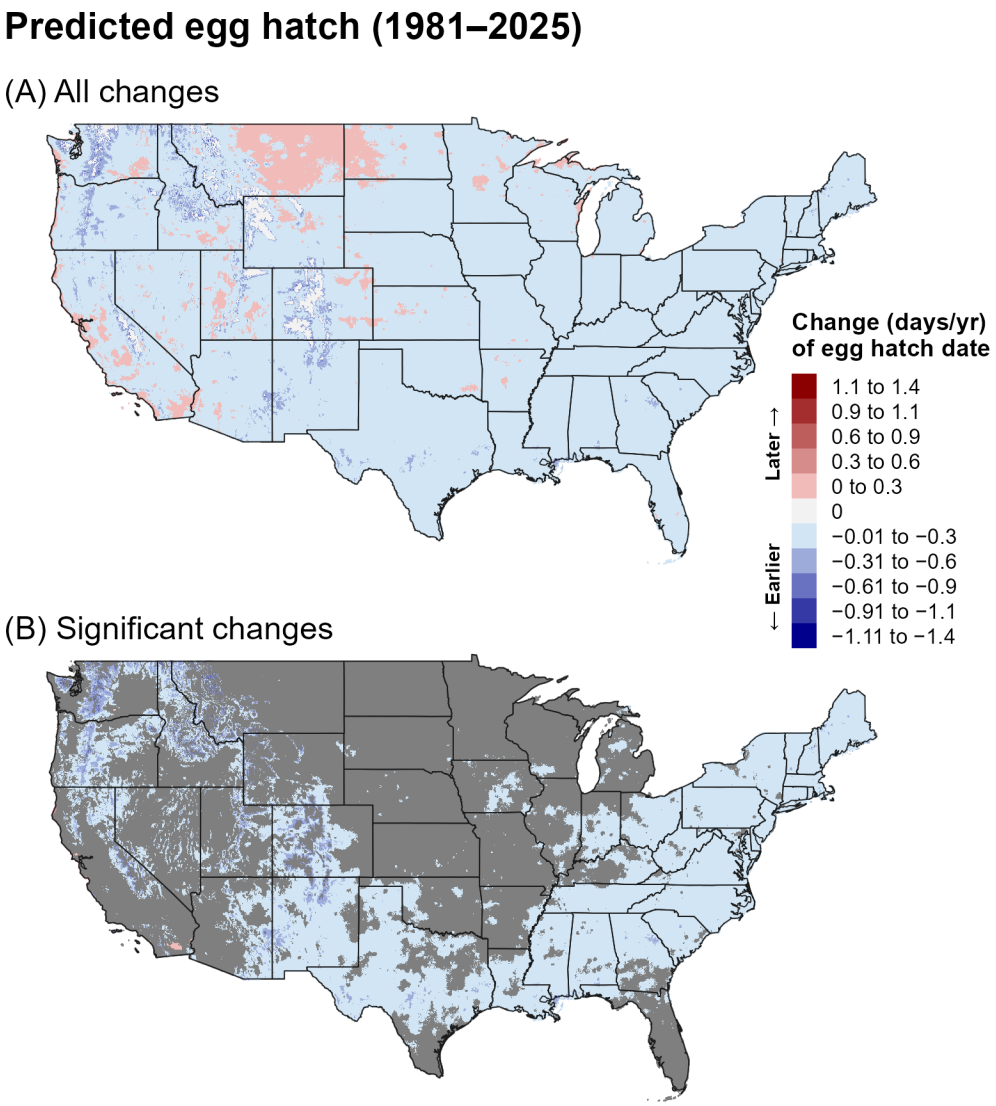


Figure 2. Changes (days/year) in predicted dates of first egg hatch for ASRB across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

Recent period: 2001–2025

Trends in predictions of first adult emergence and egg hatch for ASRB over the recent 24-year period (2001–2025) were partially consistent with those observed over the 1981–2025 period. In the eastern United States, significant trends towards earlier phenology were larger in magnitude but less spatially extensive (Figure 3, Figure 4), whereas most trends in the West were not significant. In contrast to the longer-term trends, isolated areas of the interior West, primarily in the southern Intermountain West, showed trends toward later phenological events.

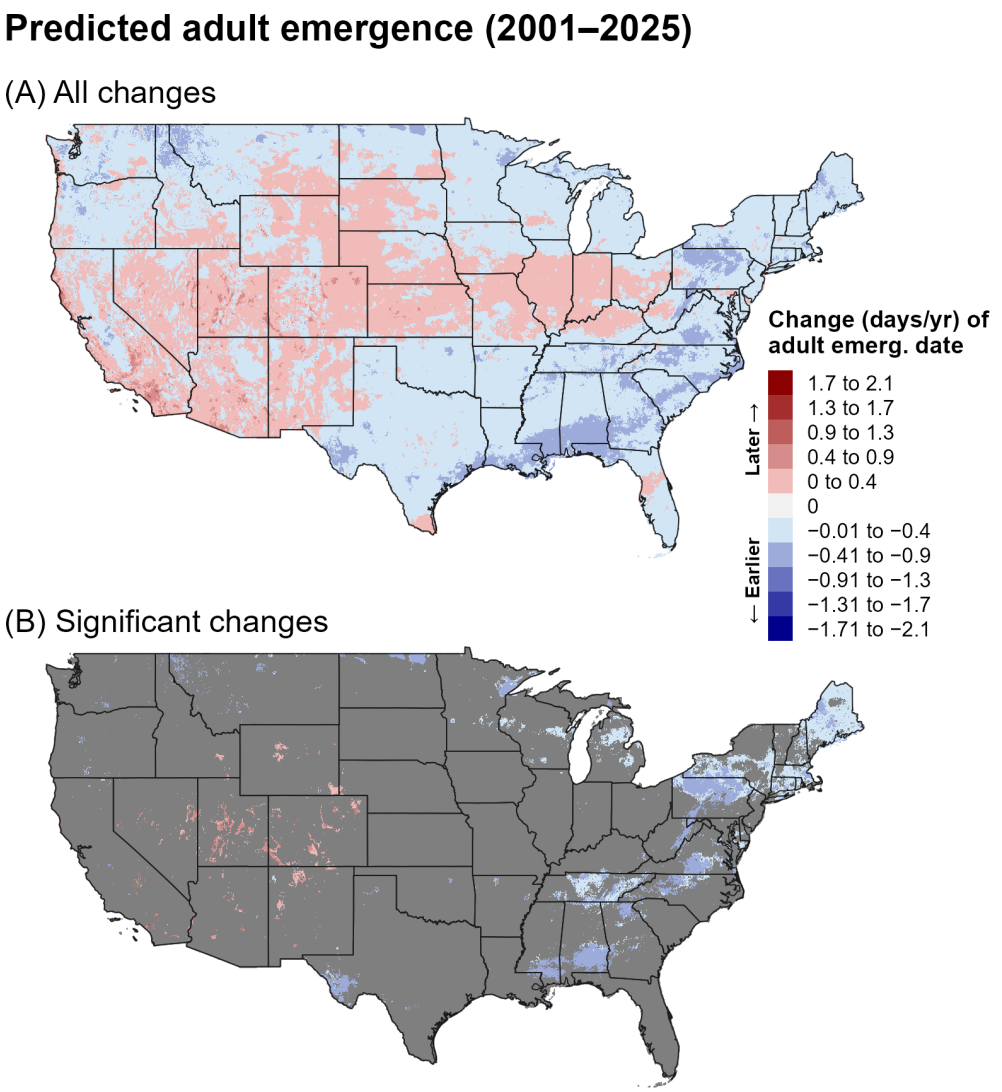


Figure 3. Changes (days/year) in predicted dates of first adult emergence for ASRB across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

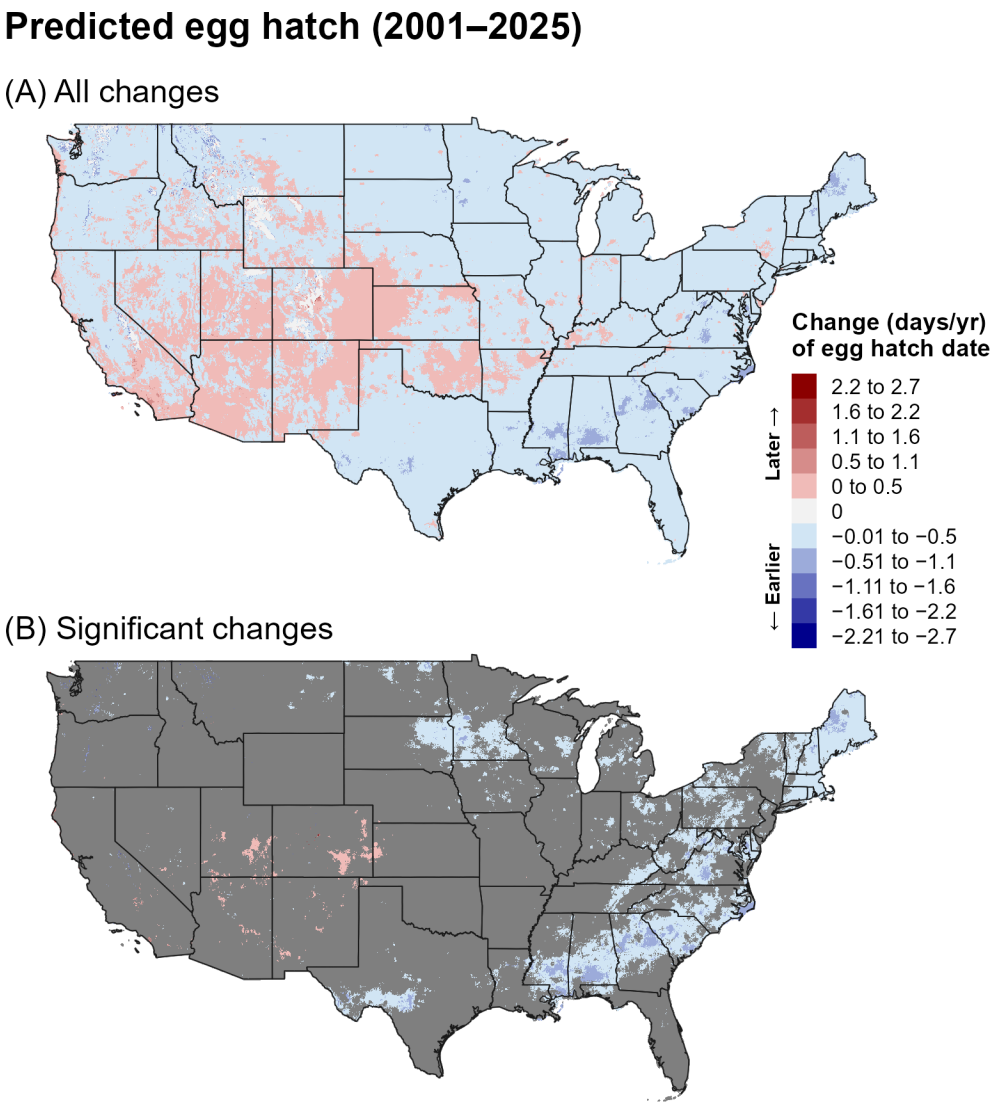


Figure 4. Changes (days/year) in predicted dates of first egg hatch for ASRB across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant.

ii. Climatic suitability

Entire period: 1981–2025

Over the entire 44-year period (1981–2025), predicted cold stress for ASRB decreased significantly across the northern United States, including portions of the Great Lakes region and northeastern United States, as well as parts of the Rocky Mountains (Figure 5). When annual accumulated stress units exceed limits specified in DDRP models, climatic suitability may change. However, changes in stress units for ASRB did not correspond to widespread shifts in suitability; only a small area of Arizona showed a significant decrease in predicted suitability (Figure 7).

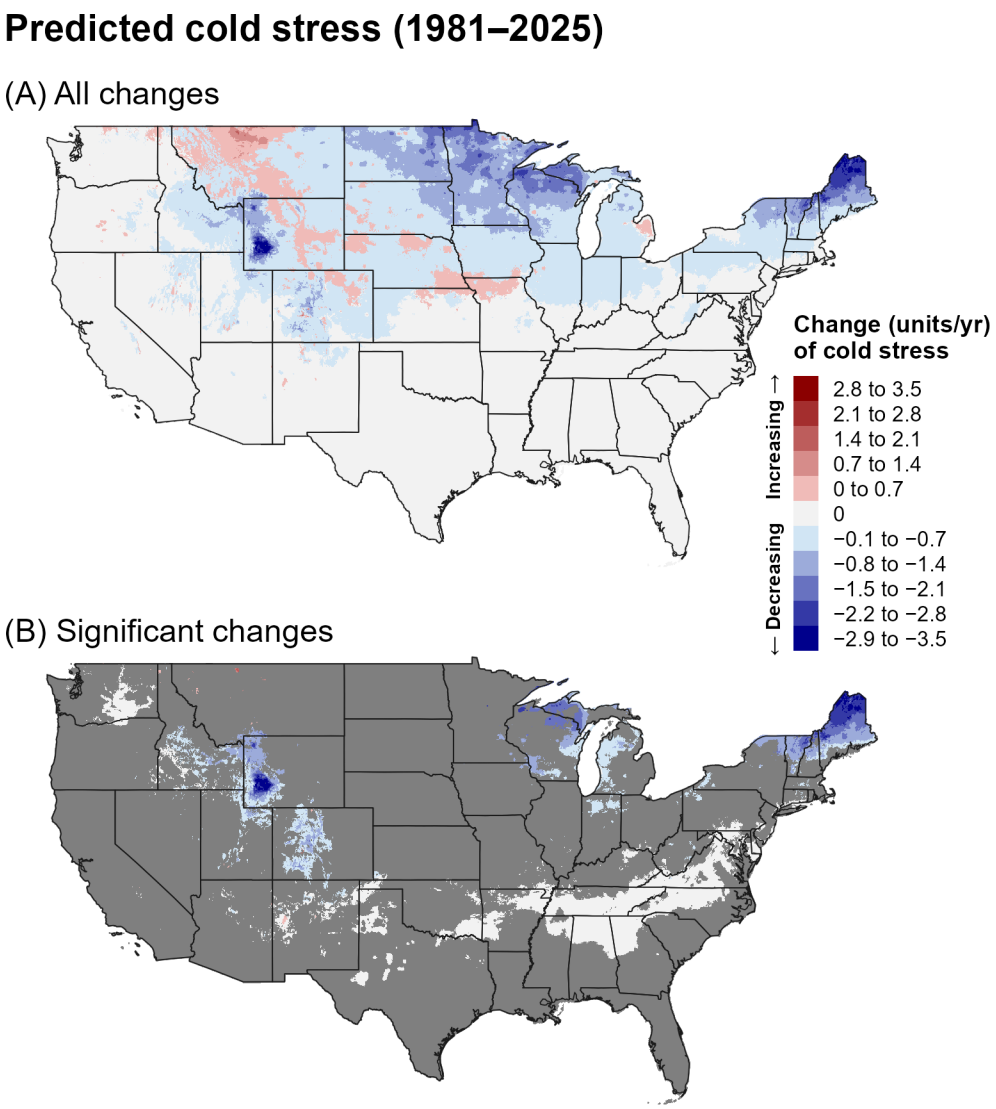


Figure 5. Changes (units/year) in predicted cold stress for ASRB across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

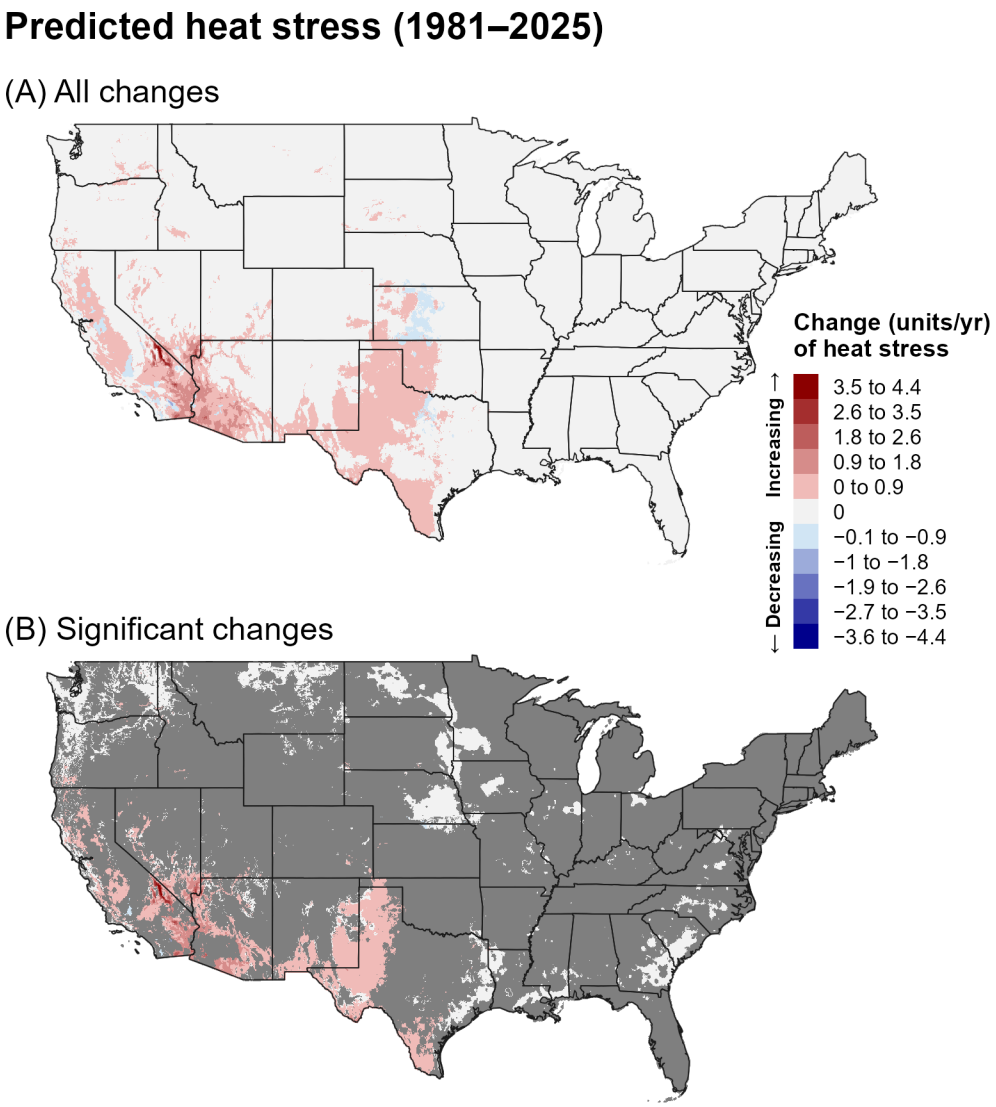


Figure 6. Changes (units/year) in predicted heat stress for ASRB across the CONUS over the entire 44-year period (1981–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

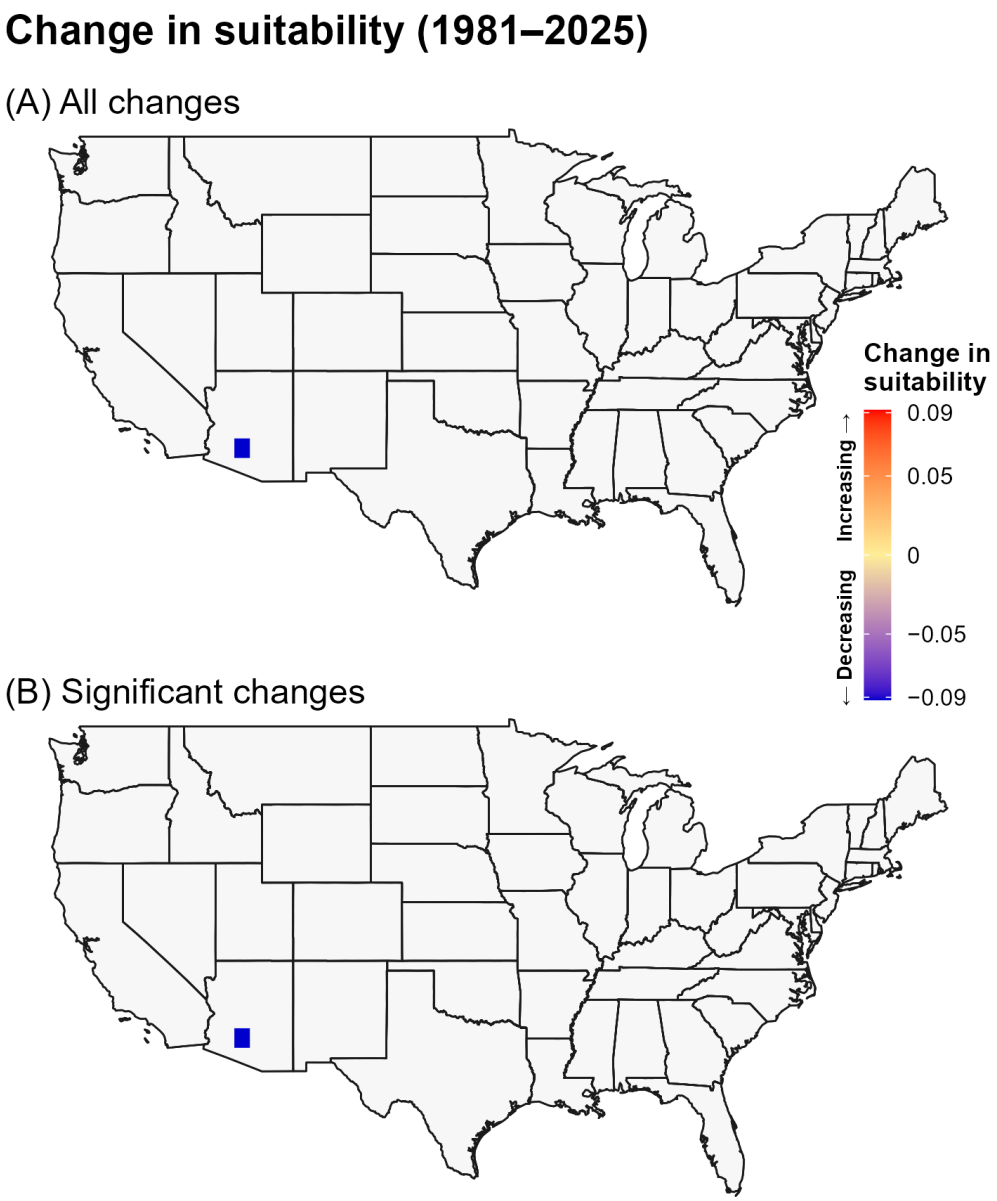


Figure 7. Changes in predicted climatic suitability for ASRB across the CONUS over the 44-year period (1981–2025). Suitability is defined as areas not excluded by cold and heat stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

Recent period: 2001–2025

Many areas that exhibited significant decreases in cold stress for ASRB over 1981–2025 no longer showed significant trends during the recent period, while significant increases in predicted cold stress emerged across portions of the northern and central Great Plains (Figure 8). Significant increases in heat stress were more localized to the Southwest, whereas parts of California showed significant decreases (Figure 9). As in the 1981–2025 analysis, significant changes in climatic suitability were limited to a small area of Arizona (Figure 10), indicating that predicted establishment suitability remained relatively stable across most of the CONUS.

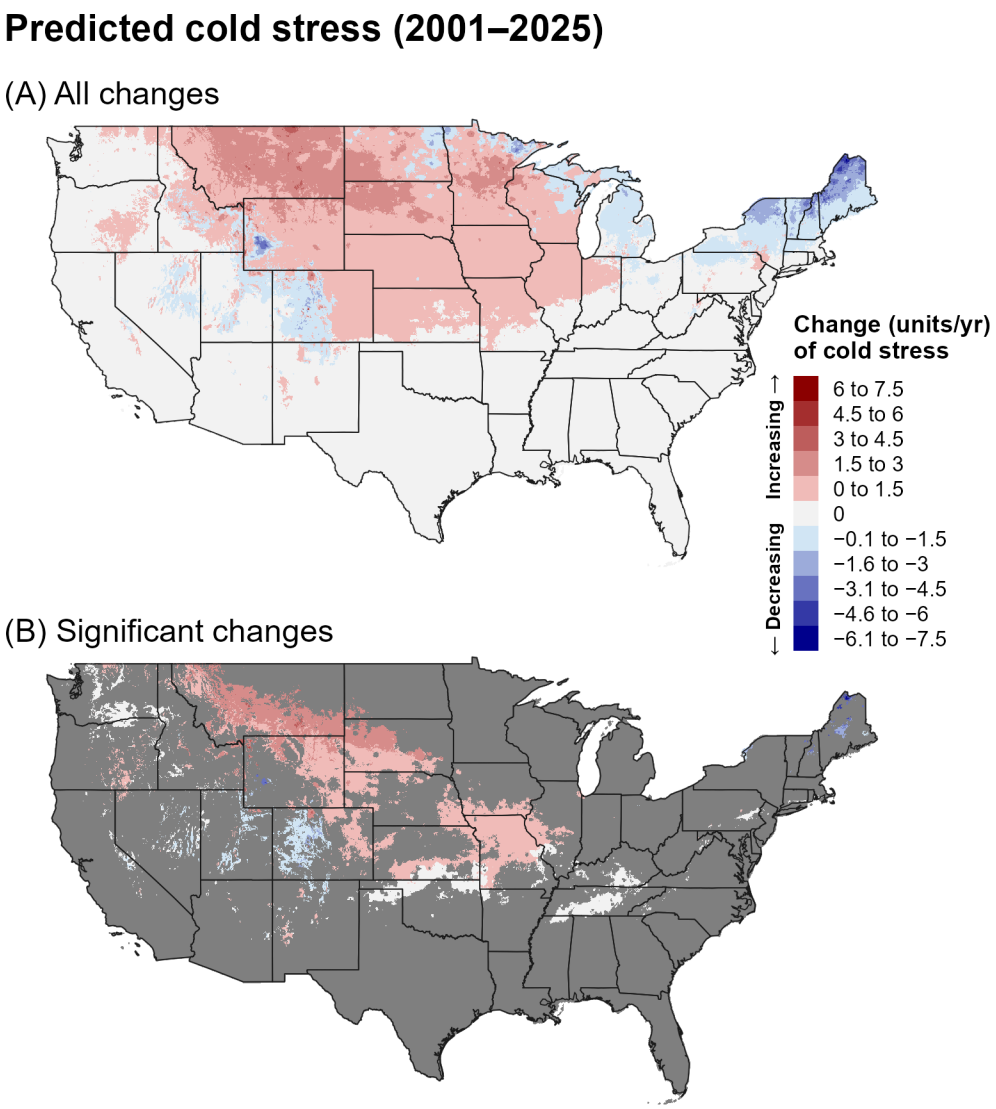


Figure 8. Changes (units/year) in predicted cold stress for ASRB across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

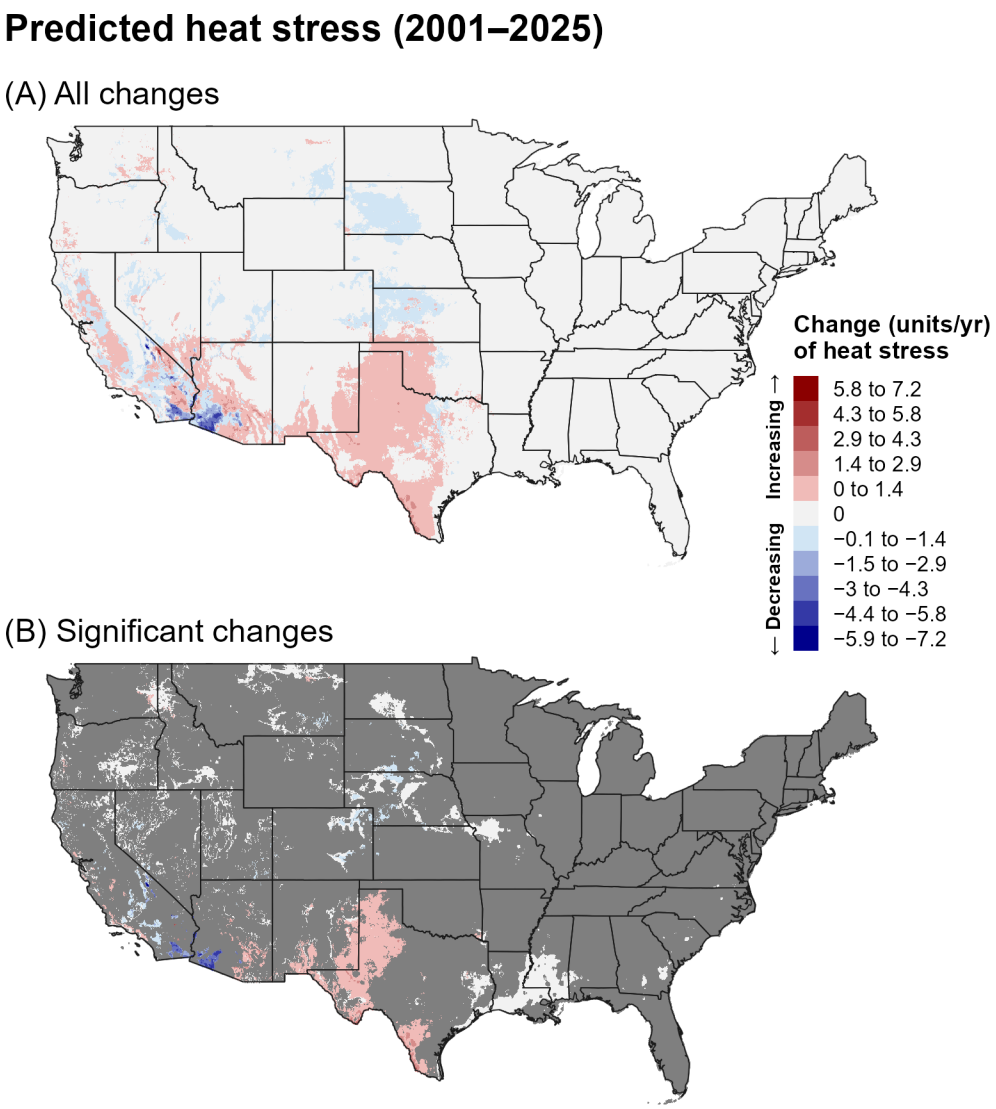


Figure 9. Changes (units/year) in predicted heat stress for ASRB across the CONUS over the recent 24-year period (2001–2025). Maps show (A) all changes and (B) statistically significant changes ($P < 0.1$); areas shown in gray are not statistically significant. Areas shown as 0 in (B) had significant trends but very small changes.

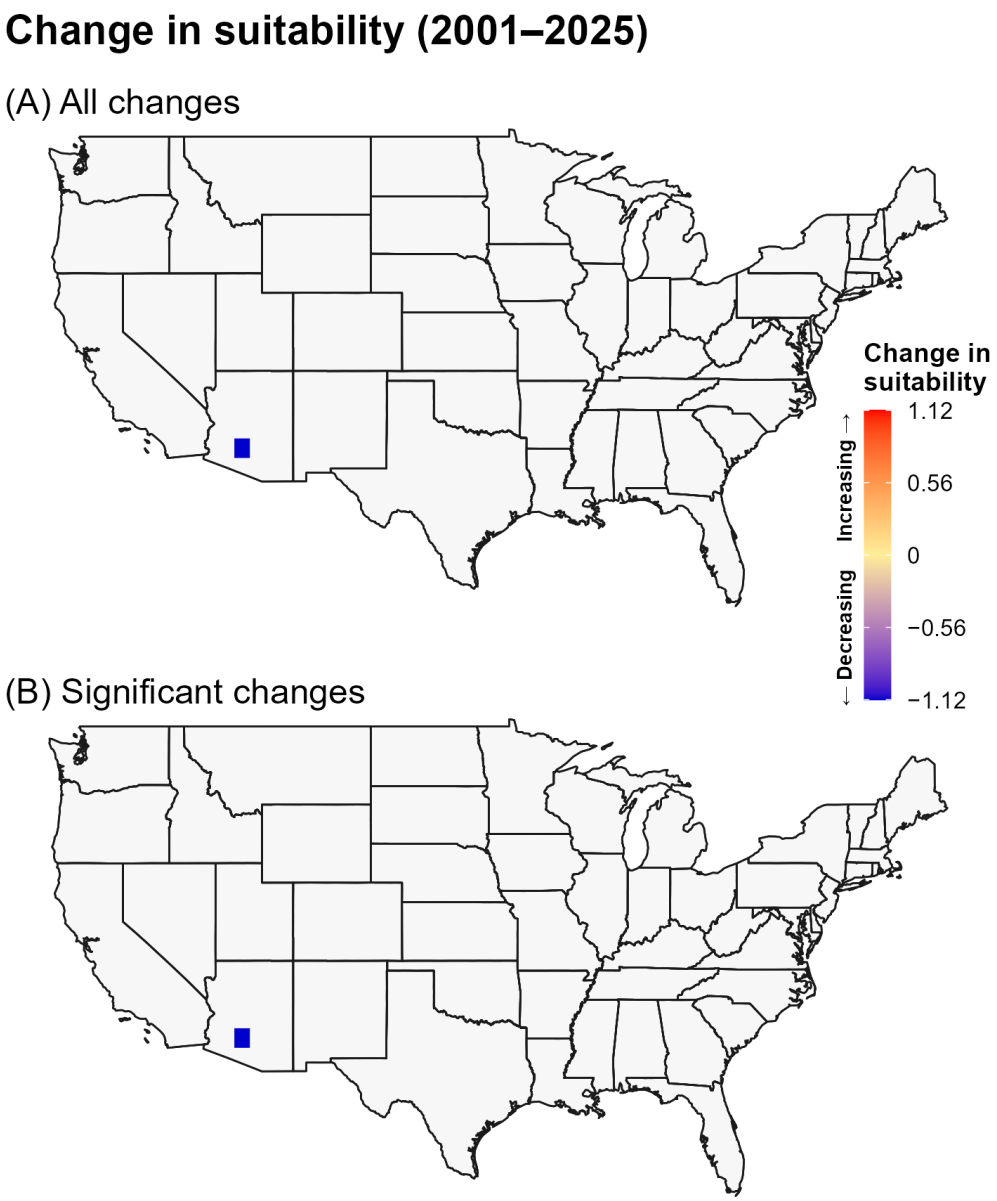


Figure 10. Changes in predicted climatic suitability for ASRB across the CONUS over the recent 24-year period (2001–2025). Suitability is defined as areas not excluded by cold and heat stress. Positive changes (warmer colors) indicate an increased likelihood of establishment, whereas negative changes (cooler colors) indicate a decreased likelihood of establishment.

2C. Discussion

Although ASRB is not established in the CONUS, understanding predicted changes in phenology can inform surveillance planning. In areas with earlier phenological events, predicted dates shifted earlier by approximately 0.01–0.5 days per year, suggesting that surveillance materials may need to be deployed earlier in the season. The southern United States warrants particular attention because it includes major rice-producing states, such as Louisiana, Texas, and Arkansas, where surveillance could help detect potential introductions into areas with suitable host crops.

Although warmer growing-season temperatures can advance ASRB emergence and increase voltinism,

temperature effects on development and overwintering are context dependent. ASRB is univoltine in cooler parts of its native range, such as northern Japan, where shortening day lengths serve as the primary environmental cue for larvae to enter winter diapause. In contrast, populations in warmer regions, including Taiwan, can complete 5–6 generations annually and typically lack diapause (Kiritani and Iwao 1967, Ramoneda and Roig 1993, Tsumuki et al. 1994). Genetic differentiation among ASRB populations in China is also associated with geographic variation in voltinism (Meng et al. 2008). Warmer growing-season temperatures have been linked to earlier emergence and spring flight, increased voltinism (Lee 2014), and greater pupal weight and fecundity (Fu et al. 2016, Huang et al. 2018). However, experimental studies found that warmer winter temperatures (10 °C versus 1 and 5 °C) reduced pupation rates, pupal weight, and post-diapause reproductive potential (Xiao et al. 2017). Collectively, these findings suggest that the effects of temperature on ASRB development and population dynamics depend on seasonal conditions and geographic context, underscoring the importance of considering both phenological shifts and climatic suitability when prioritizing surveillance.

The climate-stress maps showed regional changes in both cold and heat stress, but these changes produced little statistically significant change in predicted climatic suitability. This distinction is important because changes in individual stress variables do not necessarily translate into changes in establishment likelihood when the modeled conditions remain within the suitable range. Decreasing cold stress in portions of the northern United States and could reduce a temperature-related limitation on establishment, whereas increasing heat stress in parts of the Southwest and California could impose greater climatic constraints. However, because the climatic-suitability maps showed very few significant changes, these stress trends did not translate into widespread changes in predicted establishment likelihood.

The stronger and more spatially extensive trends in predicted phenology and climate stress over the entire 44-year period may reflect both the larger sample size, which can reduce noise and improve detection of directional patterns, and differences in the underlying climate trends between the two time periods. In particular, the recent 24-year period may capture climate variability or regional trends that differ from those observed over the longer 44-year period. Thus, differences between the two periods should be interpreted as reflecting both long-term climate trends and shorter-term regional variability.

2D. References

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